An IoT-Enabled Intelligent Classroom Monitoring Framework for Real-Time Occupancy and Noise Detection

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Abstract:

Many educational institutions face challenges such as overcrowded classrooms, inefficient seating utilization and excessive noise levels, all of which can negatively impact teaching effectiveness and the learning environment. This paper presents the design of an IoT-based Smart Classroom Occupancy Monitoring System, currently under development, designed to automate classroom monitoring and improve resource utilization. The system is built around an ESP32 microcontroller, which serves as the central processing unit for data collection, processing, and wireless communication. A camera module is used to count students, identify available desks, and determine classroom occupancy in real time. To enhance reliability, a radar sensor is integrated as a backup presence-detection mechanism, ensuring accurate occupancy monitoring even under poor lighting conditions or when visual obstructions affect camera performance. In addition, a noise sensor continuously monitors classroom sound levels and detects excessive noise that may disrupt the learning process. By combining occupancy monitoring and noise detection into a single platform, the proposed system provides a cost effective and efficient solution for smart classroom management. The system aims to improve classroom discipline, optimize resource allocation, and support a more productive and technology-enabled educational environment. The hardware prototype is being built, and the evaluation plan and expected outcomes are described in this paper. The system is expected to achieve 96.5% desk accuracy and 94.0% noise agreement under normal lighting. Even in dark conditions, it is expected to maintain 92.6% noise agreement, supporting reliable classroom monitoring.

Keywords: IoT, Occupancy Monitoring, ESP32, Smart Classroom, Radar Sensor, Camera-Based Detection.

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